Chapter 1.3_Separable ODEs

Example-1

Solve the ODE $y' = 1 + y^2$

Solution:

$y' = 1 + y^2$

$$or, \quad \frac{dy}{dx} = 1 + y^2$$

$$or, \quad \frac{1}{1+y^2}dy = dx$$

Integrating we get,

$$or, \quad \int \frac{1}{1+y^2}dy = \int dx$$

$$or, \quad tan^{-1}y = x + c$$

$$or, \quad y = \tan(x + c)$$

Example-2

Solve the ODE $y' = (x+1)e^{-x}y^2$

Solution:

$y' = (x+1)e^{-x}y^2$

$$or, \quad \frac{dy}{dx} = (x+1)e^{-x}y^2$$

$$or, \quad \frac{1}{y^2}dy = (x+1)e^{-x}dx$$

Integrating we get,

$$or, \quad \int \frac{1}{y^2}dy = \int (x+1)e^{-x}dx$$

$$or, -\frac{1}{y} = (x+1)\int e^{-x}dx - \int \left\{\frac{d}{dx}(x+1)\int e^{-x}dx\right\}dx$$

$$or, -\frac{1}{y} = -(x+1)e^{-x} + \int e^{-x}dx\,dx$$

$$or, -\frac{1}{y} = -(x+1)e^{-x} - e^{-x} + c$$

$$or, -\frac{1}{y} = -(x+2)e^{-x} + c$$

$$or, \quad \frac{1}{y} = (x+2)e^{-x} - c$$

CamScanner

$$or, \quad y = \frac{1}{(x+2)e^{-x} - c}$$

This is the general solution of the given differential equation.

Problem set:

2. **Find a general solution of the differential equation $y^3y' + x^3 = 0$**

Solution:

$y^3y' + x^3 = 0$

$$or, \quad y^3\frac{dy}{dx} = x^3$$

$$or, \quad y^3dy = x^3dx$$

Integrating we get,

$$or, \quad \int y^3dy = \int x^3dx$$

$$or, \quad \frac{y^4}{4} = \frac{x^4}{4} + c_1$$

$$or, \quad y^4 = x^4 + 4c_1$$

$$or, \quad y^4 = x^4 + c$$

This is the general solution of the given differential equation.

3. **Find a general solution of the differential equation $y' = sec^2y$**

Solution: Given differential equation is

$y' = sec^2y$

$$or, \quad \frac{dy}{dx} = sec^2y$$

$$or, \quad cos^2ydy = dx$$

$$or, \quad \frac{1}{2} \times 2cos^2ydy = dx$$

$$or, \quad \frac{1}{2}(1 + cos2y)dy = dx$$

$$or, \quad \int (1 + cos2y)dy = 2\int dx$$

$$or, \quad y + \frac{1}{2}sin2y = 2x + c$$

This is the general solution of the given differential equation.

4. **Find a general solution of the differential equation**

$$y'sin2\pi x = \pi ycos2\pi x$$

Solution: Given differential equation is

$y' sin2\pi x = \pi y cos2\pi x$

or, $\frac{1}{y}\frac{dy}{dx} = \pi \frac{cos2\pi x}{sin2\pi x}$

or, $\int \frac{1}{y} dy = \pi \int \frac{cos2\pi x}{sin2\pi x} dx$

or, $lny = \pi \ln(sin2\pi x) \times \frac{1}{2\pi} + lnc_1$

or, $2lny = \ln(sin2\pi x) + 2lnc_1$

or, $lny^2 = \ln(sin2\pi x) + lnc_1{}^2$

or, $lny^2 = \ln c_1{}^2(sin2\pi x)$

or, $y^2 = c_1{}^2(sin2\pi x)$

or, $y^2 = csin2\pi x$, where $c_1{}^2 = c$ is an integrating constant

This is the general solution of the given differential equation.

5. Find a general solution of the differential equation

$\boldsymbol{yy' + 36x = 0}$

Solution: Given differential equation is

$yy' + 36x = 0$

or, $y\frac{dy}{dx} = -36x$

or, $ydy = -36xdx$

Integrating we get,

or, $\int ydy = -36 \int xdx$

or, $\frac{y^2}{2} = -36 \times \frac{x^2}{2} + c_1$

or, $y^2 = -36x^2 + 2c_1$

or, $36x^2 + y^2 = c$

where $2c_1 = c$ is an integrating constant

This is the general solution of the given differential equation.

6. Find a general solution of the differential equation

$\boldsymbol{y' = e^{2x-1}y^2}$

Solution: Given differential equation is

$y' = e^{2x-1}y^2$

or, $\frac{1}{y^2}dy = e^{2x-1}dx$

CamScanner

Integrating we get,

$$or, \quad \int \frac{1}{y^2} dy = \int e^{2x-1} dx$$

$$or, \quad -\frac{1}{y} = \frac{1}{2} e^{2x-1} + c$$

$$or, \quad \frac{1}{2} e^{2x-1} + \frac{1}{y} + c = 0$$

This is the general solution of the given differential equation.

7. Find a general solution of the differential equation

$$\boldsymbol{xy' = y + 2x^3 sin^2 \frac{y}{x}}$$

Solution: Given differential equation is

$$xy' = y + 2x^3 sin^2 \frac{y}{x}$$

$$or, \quad x\frac{dy}{dx} = y + 2x^3 sin^2 \frac{y}{x}$$

$$or, \quad \frac{dy}{dx} = \frac{y}{x} + 2x^2 sin^2 \frac{y}{x} - - - - - - (i)$$

Put $\frac{y}{x} = u$

$$or, \quad y = ux$$

$$or, \quad \frac{dy}{dx} = u + x\frac{du}{dx}$$

Equation (i) becomes

$$or, \quad u + x\frac{du}{dx} = u + 2x^2 sin^2 u$$

$$or, \quad x\frac{du}{dx} = 2x^2 sin^2 u$$

$$or, \quad cosec^2 u du = 2x dx$$

$$or, \quad \int cosec^2 u du = \int 2x dx$$

$$or, -cotu = x^2 + c_1$$

$$or, cot\frac{y}{x} = -x^2 - c_1$$

$or, \ cot\frac{y}{x} = c - x^2$; where $c = -c_1$ is an integrating constant

$$or, \quad \frac{y}{x} = cot^{-1}(c - x^2)$$

$$or, \quad y = xcot^{-1}(c - x^2)$$

This is the general solution of the given differential equation.

8. **Find a general solution of the differential equation**

$y' = (y + 4x)^2$

Solution: Given differential equation is

$y' = (y + 4x)^2$

$$or, \quad \frac{dy}{dx} = (y + 4x)^2 ----(i)$$

Put $y + 4x = v$

$$or, \quad \frac{dy}{dx} = \frac{dv}{dx} - 4$$

Equation (i) becomes

$$or, \quad \frac{dv}{dx} - 4 = v^2$$

$$or, \quad \frac{dv}{dx} = 4 + v^2$$

$$or, \quad \frac{1}{4 + v^2} dv = dx$$

Integrating we get,

$$\frac{1}{2} tan^{-1} \frac{v}{2} = x + c_1$$

$$tan^{-1}\left(\frac{y + 4x}{2}\right) = 2(x + c_1)$$

$$or, \quad \frac{y + 4x}{2} = tan(2x + c)$$

where $2c_1 = c$ is an integrating constant

This is the general solution of the given differential equation.

9. **Find a general solution of the differential equation**

$$xy' = y^2 + y$$

Solution: Given differential equation is

$$xy' = y^2 + y$$

$$or, \quad x\frac{dy}{dx} = y^2 + y ----(i)$$

Put $\frac{y}{x} = u$

$$or, \quad y = ux$$

$$or, \quad \frac{dy}{dx} = u + x\frac{du}{dx}$$

Equation (i) becomes

CamScanner

$$or, \quad x\left(u + x\frac{du}{dx}\right) = u^2x^2 + ux$$

$$or, \quad u + x\frac{du}{dx} = u^2x + u$$

$$or, \quad x\frac{du}{dx} = u^2x$$

$$or, \quad \frac{1}{u^2}du = dx$$

$$or, -\frac{1}{u} = x + c$$

$$or, \quad u = -\frac{1}{x+c}$$

$$or, \quad \frac{y}{x} = -\frac{1}{x+c}$$

$$or, \quad y = -\frac{x}{x+c}$$

This is the general solution of the given differential equation.

10. Find a general solution of the differential equation

$$\boldsymbol{xy' = x + y}$$

Solution: Given differential equation is

$$xy' = x + y$$

$$or, \quad \frac{dy}{dx} = 1 + \frac{y}{x} ----(i)$$

Put $\frac{y}{x} = u$

$$or, \quad y = ux$$

$$or, \quad \frac{dy}{dx} = u + x\frac{du}{dx}$$

Equation (i) becomes

$$or, \quad u + x\frac{du}{dx} = 1 + u$$

$$or, \quad x\frac{du}{dx} = 1$$

$$or, \quad du = \frac{1}{x}dx$$

$$or, \quad u = lnx + c$$

$$or, \quad \frac{y}{x} = lnx + c$$

This is the general solution of the given differential equation.

CamScanner

19. **Exponential growth.** If the growth rate of the number of bacteria at any time t is proportional to the number present at t and doubles in 1 week, how many bacteria can be expected after 2 weeks? After 4 weeks?

Solution:

Let, N = the number of bacteria at time t

According to question,

$\frac{dN}{dt} \propto N$

$or, \frac{dN}{dt} = kN \text{ ------}(i)$

Where k is the constant of growth.

Let N_0 be the initial number of bacteria

Then, after 1 week $N = 2N_0$

after 2 weeks $N = 4N_0$

after 3 weeks $N = 8N_0$

Integrating equation (i) we get,

$\int \frac{1}{N} dN = k \int dt$

$or, lnN = kt + lnc$

$or, lnN = lne^{kt} + lnc$

$or, lnN = lnce^{kt}$

$or, N = ce^{kt} \text{ ----}(ii)$

At $t = 0, N = N_0$

$\therefore N_0 = ce^0$

$or, c = N_0$

$\therefore N = N_0 e^{kt}$

At $t = 1, N = 2N_0$

$\therefore 2N_0 = N_0 e^k$

$or, e^k = 2$

$or, e^k = e^{ln2}$

$or, k = ln2$

$\therefore N = N_0 e^{tln2}$

Therefore, the number of bacteria at any time t is $N = N_0 e^{tln2}$, where N_0 is the number of bacteria at initial time.

After 2 weeks the number of bacteria will be $N = N_0 e^{2ln2} = 4N_0$

After 4 weeks the number of bacteria will be $N = N_0 e^{4ln2} = 2^4 N_0 = 16N_0$

CamScanner

24. **Mixing problem.** A tank contains 400 gal of brine in which 100 lb of salt are dissolved. Fresh water runs into the tank at a rate of 2 gal/min. The mixture, kept practically uniform by stirring, runs out at the same rate. How much salt will there be in the tank at the end of 1 hour?

Solution:

Volume of tank = 400 gal

Amount of salt = 100 lb

Let, y be the amount of salt in the tank at time t

Rate of change = Rate in – Rate out

At time t,

Rate in = $(2\ gal/min) \times (0\ lb/gal) = 0\ lb/min$

Rate out = $(2\ gal/min) \times \left(\frac{y}{400}\ lb/gal\right) = \frac{y}{200}\ lb/\ min$

Rate of change $\frac{dy}{dt}$ = Rate in – Rate out

$$= 0 - \frac{y}{200} lb/min = -\frac{y}{200}\ lb/\ min$$

$$\therefore \frac{dy}{dt} = -\frac{y}{200}$$

$$or, \quad \frac{1}{y}dy = -\frac{1}{200}dt$$

$$or, \quad \int \frac{1}{y}dy = -\frac{1}{200}\int dt$$

$$or, \quad lny = -\frac{1}{200}t + lnc$$

$$or, \quad lny = lne^{-\frac{t}{200}} + lnc$$

$$or, \quad lny = lnce^{-\frac{t}{200}}$$

$$or, \quad y = ce^{-\frac{t}{200}}$$

At $t = 0, y = 100$

$\therefore 100 = ce^0$

$or, \quad c = 100$

$\therefore y = 100e^{-\frac{t}{200}}$

At $t = 60,\ \ y = 100 \times e^{-\frac{60}{200}} \approx 74.08\ lb$

CamScanner